

Week 6 Project Summary

HealthConnect Project Kickoff and Problem Understanding

Experience Lab Internship Programme

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1. What I Planned to Accomplish

Per the Week 6 brief, the goal was not to rebuild the Week 5 baseline but to advance it: analyse the Week 5 model's weaknesses through error analysis, refine features where justified, develop and compare at least one improved modelling approach, complete a genuine cross-track integration activity (not just communication), and validate whether the resulting model is actually suitable for HealthConnect's operational use case.

2. What I Completed

- Reproduced the Week 5 baseline to generate predictions, then ran a structured error analysis: confusion matrix breakdown, false-positive/false-negative feature profiling, and accuracy-by-appointment-type segmentation.
- Resolved the Week 5 multicollinearity issue by dropping the redundant `previous_appointments` feature.
- Completed a genuine cross-track integration activity with the Data Analytics track (Section 5), which directly produced a new engineered feature.
- Engineered two new features: `booking_lead_days_sq` (captures the non-linear lead-time effect) and `long_lead_followup` (from the cross-track finding).
- Built and compared three modelling approaches — refined Logistic Regression, Random Forest, and Gradient Boosting — against the Week 5 baseline, using both a held-out test set and 5-fold grouped cross-validation.
- Added a business-relevance evaluation (precision/lift within the top 20% highest-risk predictions) beyond standard classification metrics.
- Documented two joint candidate models and updated risks/limitations and Week 7 testing requirements.

3. What I Improved from Week 5

- Multicollinearity between patient-history features: resolved by removing the redundant raw count.
- Model performance: refined Logistic Regression improved ROC-AUC from 0.678 to 0.686 on the held-out test set purely from better features (same model type as Week 5).
- Non-linearity handling: added a squared lead-time term so the linear model can represent the convex no-show-rate curve identified in Week 4/5 EDA.

- Diagnostic depth: Week 5 reported aggregate metrics only; Week 6 adds error-type profiling and segment-level accuracy, which surfaced two new, specific findings (the short-lead/clean-history blind spot, and the Follow-up appointment-type risk gap).

4. What I Integrated

A Data-Analytics-style segment finding (no-show rate by appointment type $\times$ booking lead time) was used as direct input to feature engineering: the discovery that Follow-up appointments show the steepest lead-time sensitivity of any appointment type motivated the `long_lead_followup` feature, which then showed measurable, non-trivial importance ($\approx 5\%$ of Random Forest feature importance) — confirming the integration produced a real, validated improvement rather than a cosmetic addition.

5. Track(s) Collaborated With

Data Analytics. Since no separate Data Analytics intern output exists in this solo run, the equivalent segment-level analysis was performed directly within the Week 6 notebook, documented transparently as a stand-in for that exchange (see the notebook, Section 4, for the full disclosure of this constraint).

6. What Was Exchanged

- Received (simulated): a no-show rate breakdown by appointment_type $\times$ booking_lead_days bucket, of the kind the Data Analytics track's Week 6 brief specifically asks them to produce and share with Data Science.
- Provided (for their advanced analytics report): the finding that Follow-up appointments are disproportionately lead-time-sensitive, along with the model-based confirmation that this segment effect carries real predictive weight — a candidate insight for their business recommendations.

7. What Changed as a Result

A new feature (`long_lead_followup`) was added to the Week 6 model specifically because of this finding, and its contribution was measured directly rather than assumed — this is the concrete evidence of integration the Week 6 brief requires, beyond simple communication between tracks.

8. Key Findings and Development Outcomes

- Refined Logistic Regression and Random Forest are statistically indistinguishable once cross-validated (mean ROC-AUC ≈ 0.680 vs. ≈ 0.679); the Random Forest's apparent edge on a single test split was within normal fold-to-fold noise.
- Gradient Boosting underperformed both, consistently, across every cross-validation fold — a genuine, honestly-reported negative result rather than a tuned comparison.
- Business-relevance check: prioritising the top 20% highest-risk predicted appointments captures a no-show rate of $\sim 72\text{--}73\%$ versus a 50% baseline ($\sim 1.44\text{--}1.45\times$ lift) — confirms the model is operationally useful even though overall accuracy is moderate.

- A structural blind spot was identified: no-shows with short lead time and a clean prior history are effectively undetectable with the current feature set.

9. Major Challenges

- Distinguishing a genuine model improvement from noise — the single-split comparison suggested Random Forest was better, but grouped cross-validation showed that difference wasn't reliable. Reporting this honestly, rather than picking the more flattering single-split number, was a deliberate methodological decision.
- Simulating the cross-track integration in a way that produces real evidence (an actual feature, actually validated) rather than a superficial mention of “collaboration.”

10. Important Decisions

- Chose to drop `previous_appointments` rather than `previous_no_shows` or `previous_no_show_rate`, since it was the most redundant of the three given the other two remained available.
- Chose not to declare a single “winner” model, since Logistic Regression v2 and Random Forest were statistically tied — both are carried forward as joint candidates, prioritising honesty over a tidier-sounding narrative.
- Chose not to pursue Gradient Boosting tuning this week, to avoid scope creep beyond the Week 6 brief's focus on validated improvement rather than an open-ended model search.

11. Remaining Issues

- Fairness/bias review across gender and age subgroups — still outstanding, now a priority for Week 7 given a stable feature set exists to test against.
- The short-lead-time/clean-history false-negative blind spot has no clear feature-based solution yet.
- Formal Data Dictionary cross-check remains outstanding since Week 4 (low risk, but not yet closed).
- Gradient Boosting has not been tuned, so it isn't yet possible to say whether its underperformance is a genuine ceiling or a tuning artefact.

12. Contribution to the Overall HealthConnect Project

This week's work moves the Data Science track from “a baseline exists” to “a validated, improved candidate model exists, with a concrete integration point to Data Analytics and a defined hand-off to Machine Learning Engineering.” The business-relevance framing (Section 8) gives Project Management and the Data Analytics track a concrete, non-technical number — a $\sim 1.44x$ lift in top-risk predictions — that can be used directly in HealthConnect-facing recommendations, rather than an abstract accuracy figure.

13. Proposed Focus for Week 7

- Run a fairness/bias check across gender and age subgroups on both candidate models.
- Convert predicted probabilities into an operational Low/Medium/High risk tier and test its stability across folds.
- Investigate whether any additional (currently unavailable) data could address the short-lead-time/clean-history blind spot.
- Support the Machine Learning Engineering track's integration of the finalised feature set and candidate models into their pipeline.
- Optionally revisit Gradient Boosting with proper hyperparameter tuning to confirm Week 6's negative result.